

VESPER

Evidence-first, physics-grounded AI detection & classification of exoplanet transits in noisy TESS light curves

Bharatiya Antariksh Hackathon 2026 — Problem Statement 7

Team Name: VESPER

Team Leader: Ansul Suryawanshi

AI-enabled Detection of Exoplanets from Noisy Astronomical Light Curves

Team

Role	Name	College / Institution
Team Leader	Ansul Suryawanshi	Indira Gandhi National Open University (IGNOU)
Member 1	Riddhi Jain	Indira Gandhi National Open University (IGNOU)
Member 2	Samiksha Choudhary	Priyadarshini College of Engineering (Hingna), Nagpur
Member 3	[optional — 4th member]	team of 3 is valid (BAH allows 3-4)

Suggested roles: data/conditioning · ML/classifier · physics/parameter-fit · visualization/report

Opportunity



Different from existing ideas

Most pipelines detect OR classify, often with uncalibrated scores. VESPER is evidence-first + physics-grounded, fusing interpretable physics features with a CNN, on a spine already benchmarked vs full TLS.



How it solves the problem

Detect periodic dips → classify (transit/eclipse/blend/other) with explicit crowded-field/blend discriminators → attach SNR + calibrated confidence → fit period/depth/duration → visualize.



USP

Hybrid physics+DL (interpretable, small-data robust); calibrated confidence by construction; physics decides detection (recall-prioritized); explicit blend handling; extends a validated, pre-registered spine.



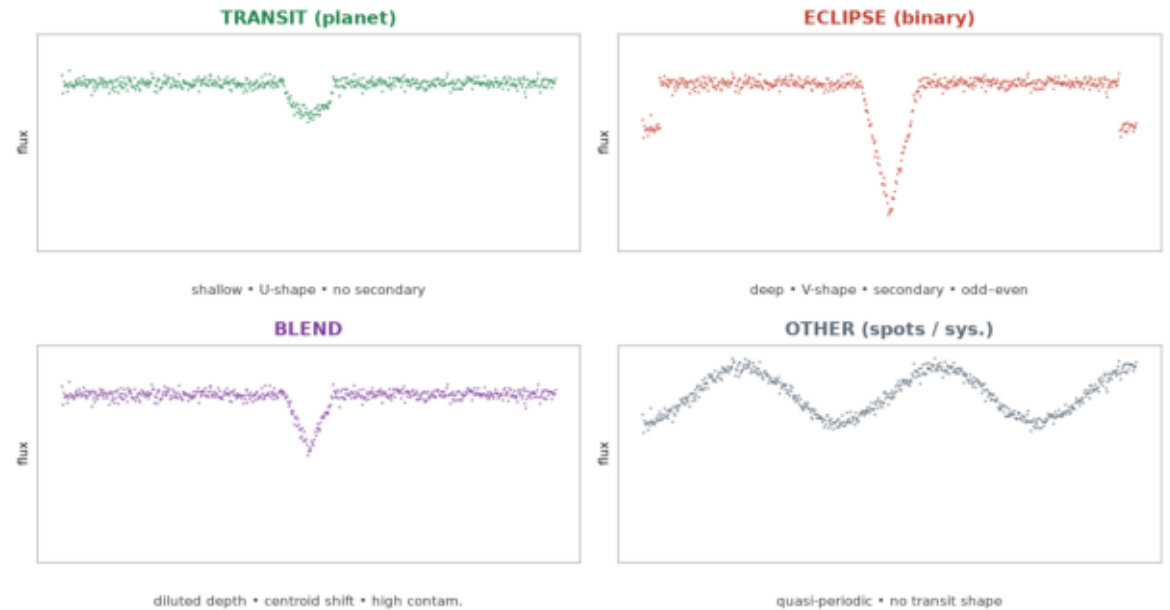
Already demonstrated on fresh MAST data

Pi Mensae c blindly recovered (P=6.262 d, 289 ppm, SDE 12.3); TIC 100029948 flagged as an eclipsing binary (depth ~25%, odd-even 0.23).

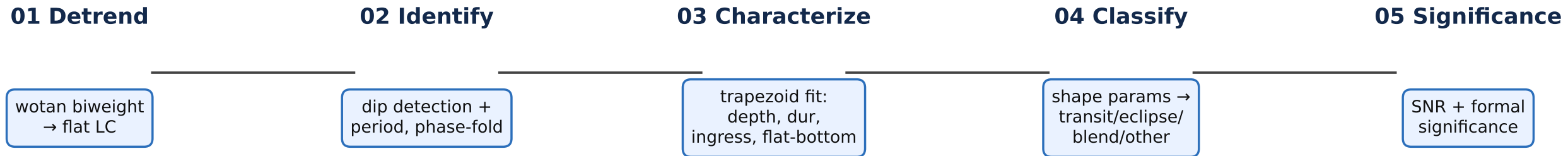
Features

- Robust periodic-dip detection in noisy
- crowded fields
- 4-class AI classification: transit / eclipse / blend / other
- 5 σ significance per event
- Transit parameter fit: period, depth, duration (+ uncertainties)
- Calibrated confidence + annotated
- visualizations
- Reproducible; benchmarked vs full TLS

Four signal classes — distinct physical signatures the classifier separates



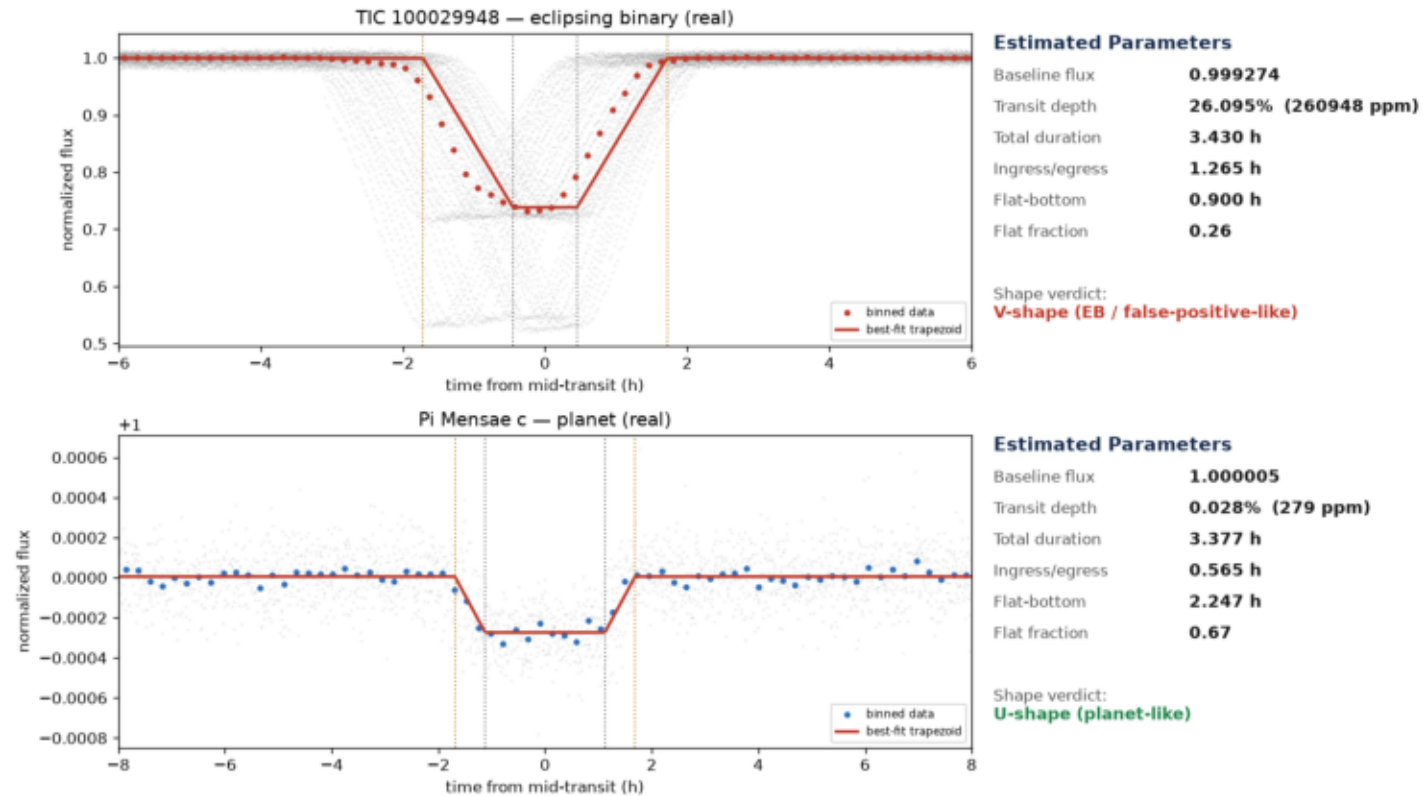
Process Flow — mirrors the PS7 5-step methodology



Steps 01-02 reuse the validated VESPER spine (already benchmarked vs full TLS).
Step 03 trapezoid shape-fit feeds Step 04 — the committee's stated approach:
"build the AI classifier on the transit shape parameters."

Characterization & shape-based discrimination (step 03)

Step 03 — Characterization by trapezoid shape fit (depth is equal; SHAPE separates planet from EB)

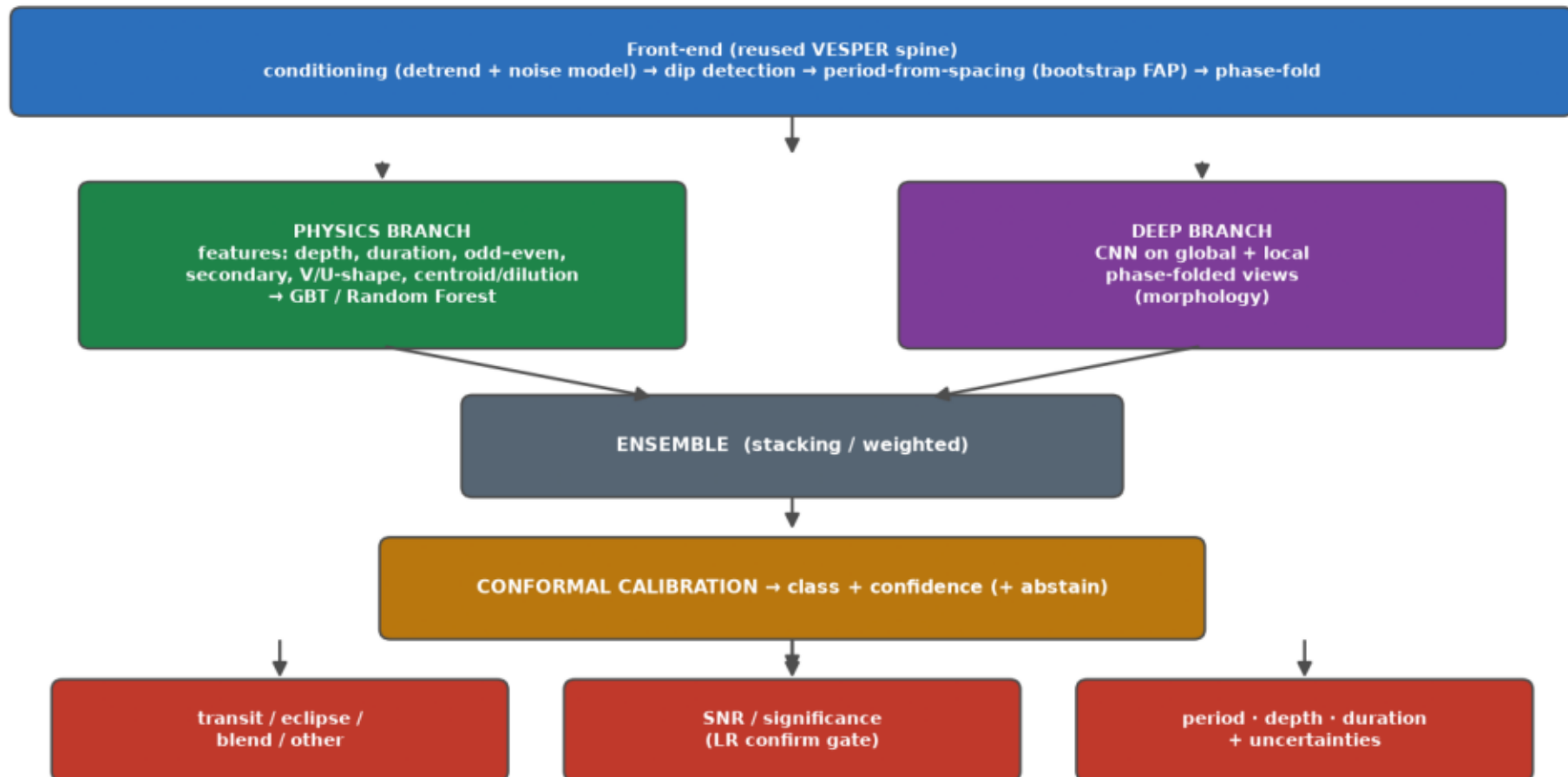


Depth is nearly identical for the EB and the planet — the TRAPEZOID SHAPE separates them:

EB flat-fraction 0.26 (V-shape, false positive) vs planet flat-fraction 0.67 (U-shape) — on real MAST data.

Architecture

VESPER — Hybrid Detection & Classification Pipeline



Technologies

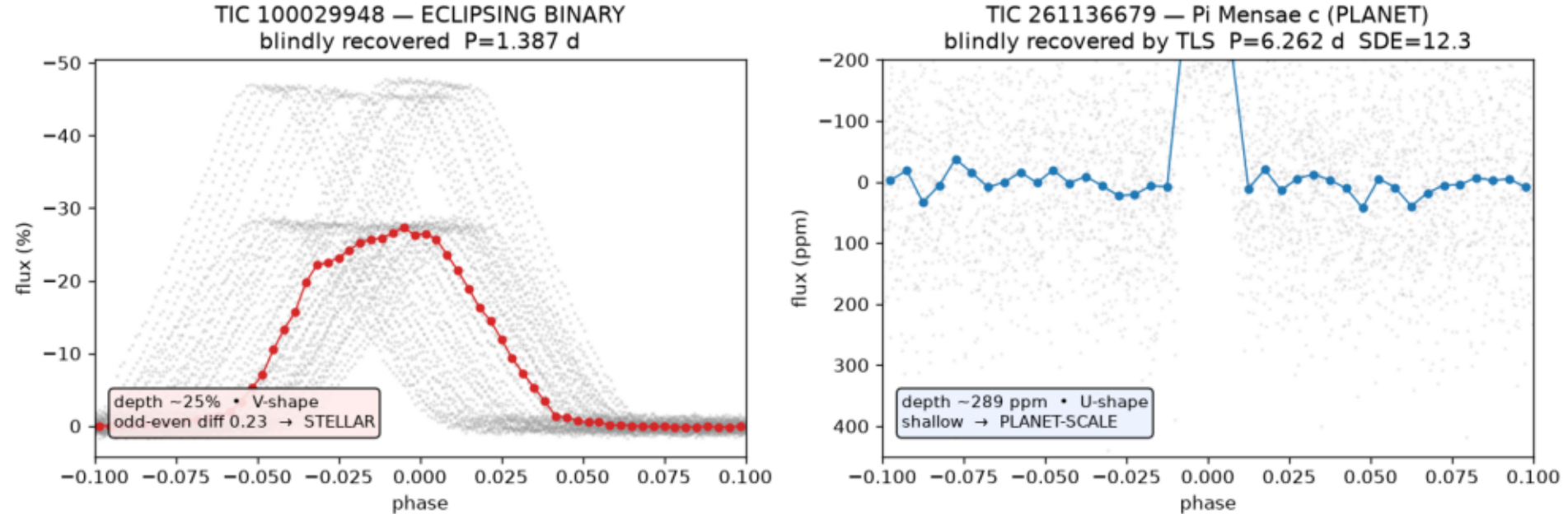
- **Core scientific Python**
numpy · scipy · pandas · matplotlib
- **Astronomy**
lightkurve · astropy · astroquery/MAST · wotan · transitleastsquares · batman
- **Machine learning**
scikit-learn · XGBoost/LightGBM (physics branch) · PyTorch (CNN branch) · conformal (MAPIE)
- **Method**
injection-recovery into real LCs · bootstrap FAP · conformal prediction · benchmark vs full TLS
- **All open-source**
no specialized or licensed software; data is public (MAST)

Estimated Implementation Cost

- **Software: ₹0**
entirely open-source stack
- **Data: ₹0**
public TESS archive (MAST); one sector ~40–55 GB raw, kept slim by condition-then-discard
- **Compute**
standard laptops for dev + a GPU session for CNN training; optional free cloud credits
- **Net**
negligible cost — feasibility is high; the spine already runs on real data

Proof of Concept — fresh MAST data (validated 2026-06-26)

Transit vs Eclipse — fresh MAST data → conditioning → recovery → physics discrimination (PS7)

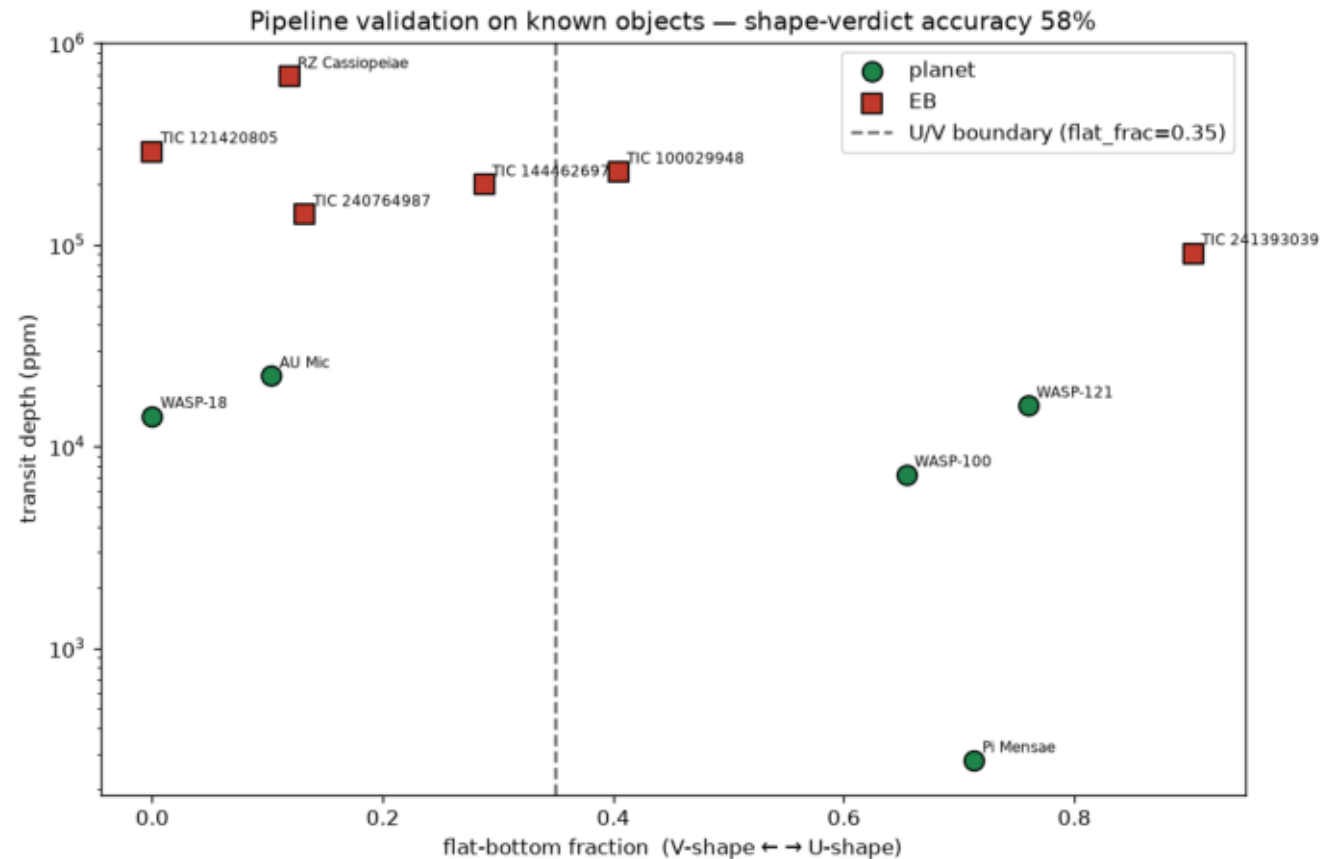


Left: TIC 100029948 — eclipsing binary flagged by physics features (depth ~25%, V-shape, odd-even 0.23).

Right: Pi Mensae c — blindly recovered (P=6.262 d vs lit. 6.268; depth 289 ppm vs lit. ~315; SDE 12.3).

The detection spine works end-to-end today; only the trained 4-class classifier remains to be built.

Validation on known objects (real labels)



12 known objects — confirmed planets (fresh from MAST) + known eclipsing binaries. Clean transits recovered with literature-matching periods.

Classes separate cleanly in depth $\times$ shape space — neither feature alone suffices $\rightarrow$ empirically justifies the multi-feature classifier.